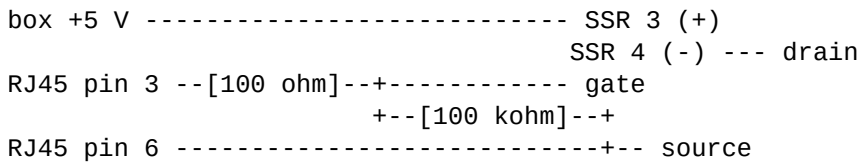
Pellevoisin: pump SSR driver

The Pi's GPIO 24 reaches the box at 3.1 V, too little to fire the SSR. A logic-level MOSFET beside the SSR switches the box's 5 V into it instead. GPIO high = pump on, as before.

Wiring

From	Through	To
box +5 V		SSR terminal 3 (+)
SSR terminal 4 (-)		MOSFET drain
RJ45 pin 3 (GPIO 24)	100 ohm	MOSFET gate
MOSFET gate	100 kohm	MOSFET source
MOSFET source		RJ45 pin 6 (5 V return = Pi GND)



Pinouts

MOSFET	Hold it	Pins, left to right
2N7000 (TO-92)	flat face towards you, legs down	source - gate - drain
RFP30N06LE (TO-220)	printed face towards you, legs down	gate - drain - source

Check

Pump commanded on: measure DC across SSR terminals 3 and 4. **4.8 V or more** = well driven. Noticeably lower with the 2N7000 = that part's threshold is high; fit the RFP30N06LE.

Never put the 12 V on the SSR in this circuit. The MOSFET's source is on the Pi's ground; the 12 V supply's negative is the organ's, and 12 V here would join the two grounds.

SSR terminals 1 and 2 (mains) are interchangeable. The 100 kohm keeps the pump off while the Pi boots.